VPN research material, part: 05

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Dive deep into this research and do comprehenisve refining! VPS hosted service has to be VPN gateway to network(s) connected to it. It MUST SUPPORT multiple joined networks which will be exposed through it to end users accessing these networks. So basically it establishes two ways VPN connection! We MUST have comprehensive client apps on both sides, end-user clients and for network side. Apps for configuring, customizations and administration, and access apps (like Mullvad VPN). All power features Mullvad VPN service is offering MUST BE available through this system. Especially connections obfuscation, especially Quic! Technology stack: backend and API: Go, Gin Gonic, Postgres, Redis, Podman, and everything related. Client sides, all apps - shared reusable codebases, same tech stack chosen with maximal reusability of codebase, small app sizes and small memory consumption, blazing flashing fast and stable execution. We MUST cutting edge solution for this. Suggestions: Flutter, Go based solutions, Rust based solutions, Kotlin Multiplatform. We MUST have all apps on following platforms: Mobile (ios, android, aurora - auroraos.ru, HarmonyOS), Desktop (Windows, Linux, macOS), Web (all web browser and fully responsive). Everything MUST event driven, real time reaponsive! The whole system!!! Rock solid, fast, stable, game changer solution!!!

Research

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Let me pick up where we left off and push this from a "working PoC" into a full-blown product-grade, Mullvad-class, multi-network VPN gateway system. The Home_VPN.md gave us the reverse-tunnel + Hysteria2/QUIC skeleton — now we need control plane, multi-network aggregation, client ecosystem, and that "game-changer" DX.

Aurora OS (Russian domestic Android replacement by Rostelecom, based on Sailfish/Android compat layer) and HarmonyOS (Huawei, Ark compiler + ability to run Flutter/ArkUI) — these two will constrain our client-stack choice, so let me search their VPN API realities quickly. Also want to align Mullvad's 2024–2025 feature list (DAITA, quantum-safe, etc.) so the "Mullvad parity" claim is grounded.